2. Utility Analysis

1. What is Utility?

Before we start, let's remember what a **"Want"** is. A want is a feeling that something is missing, and you want to fill ㄱ gap.

- **Definition of Utility:** Utility is the power of a product to satisfy a human want. In simple words, if a thing can satisfy your need, it has utility for you.
- **Daily Life Example:** When you are very thirsty, a glass of water has the "power" to quench your thirst. That power is called Utility.

2. Special Features of Utility

Utility has some unique characteristics that you should know for your exams:

- **Relative Concept:** Utility changes with time and place.
    - *Example:* Woollen clothes have more utility in winter (time) and in cold places like Kashmir (place) than in summer or in Mumbai.
- **Subjective Concept:** It differs from person to person. This is because everyone has different tastes and habits.
    - *Example:* A stethoscope has utility to a doctor, but it is of no use to a common person.
- **Ethically Neutral:** Utility doesn't care if a want is "good" or "bad." It just satisfies a want.
    - *Example:* A knife has utility to a cook to cut vegetables, but it also has utility to a criminal. Economics treats both as utility.
- **Differs from Usefulness:** A thing might have utility but might not be "useful" for your health.
    - *Example:* Liquor has utility to an addict, but it is not useful for their health.
- **Differs from Pleasure:** You might get utility from something even if it is painful.
    - *Example:* An injection has utility because it cures you, but it certainly doesn't give you any "pleasure".
- **Differs from Satisfaction:** Utility is the start (why you buy it), and satisfaction is the end (how you feel after using it).
- **Measurement is Hypothetical:** You cannot actually measure utility in numbers like 1kg or 2 liters. It is just a feeling.
- **Multi-purpose:** One thing can be used for many tasks.
    - *Example:* Electricity can be used for lights, fans, washing machines, and by many people at once.
- **Depends on Intensity of Want:** The more desperate you are for something, the higher its utility.

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- *Example:* A hungry person feels more utility in food than someone who just ate.

- **Basis of Demand:** You only demand a product if it has utility for you.

3. Types of Utility

How is utility created? There are six main ways:

1. **Form Utility:** When you change the shape or structure of something.
   - *Example:* Changing a log of wood into a table or chair.
2. **Place Utility:** When utility increases by changing the location of a good.
   - *Example:* Mangoes have more utility in the city market than at the farm. Transport creates place utility.
3. **Service Utility:** When professionals provide services to you.
   - *Example:* Services of doctors, teachers, or lawyers.
4. **Knowledge Utility:** When you learn how to use a product.
   - *Example:* A mobile phone has more utility once you learn how to use all its apps.
5. **Possession Utility:** When ownership is transferred from one person to another.
   - *Example:* When you buy a book from a shopkeeper, the utility increases because you now own it.
6. **Time Utility:** When utility changes with the time of use.
   - *Example:* A student has more utility for textbooks during exams than during vacations. Storing blood in a blood bank for emergencies also creates time utility.

4. Concepts of Utility (The Formulas)

There are two main ways to look at utility:

A. Total Utility (TU)

This is the grand total of satisfaction you get from consuming all units of a commodity.

- **Formula:**

$$TU = \sum MU$$

(Sum of all Marginal Utilities)

- **Or:**

$$TU = MU_1 + MU_2 + MU_3 ... + MU_n$$

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B. Marginal Utility (MU)

This is the extra satisfaction you get from consuming one more unit of a commodity.

- **Formula:**

$$MU_n = TU_n - TU_{(n-1)}$$

  - $$MU_n$$

: Marginal utility of the

$$n^{th}$$

unit.

  - $$TU_n$$

: Total utility at the current level.

  - $$TU_{(n-1)}$$

: Total utility at the previous level.

Daily Life Example: If the 1st mango gives you 10 points of joy and the 2nd mango gives you 8 points, your Total Utility is 18 ($10 + 8$), but your Marginal Utility for the 2nd mango is only 8.

5. Relationship between TU and MU

As you consume more units of a product, TU and MU behave differently:

1. **Start:** For the very first unit, TU and MU are the same.
2. **Growth:** As you eat more, TU increases, but it increases slowly because MU is falling.
3. **The Peak (Point of Satiety):** Eventually, TU reaches its maximum point. At this exact moment, MU is zero. You are perfectly satisfied and don't want any more.
4. **The Fall:** If you keep consuming beyond this point, TU starts to fall, and MU becomes negative (this is called "disutility" or dissatisfaction).

Exam Tip: Always remember that at the Point of Satiety, TU is highest and MU is 0.

6. The Law of Diminishing Marginal Utility (DMU)

This is the most important part of the chapter!

- **Who proposed it?** First by Prof. Gossen, but later explained in detail by Prof. Alfred Marshall in 1890.

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- **What does the law say?** "Other things remaining constant, the additional benefit which a person derives from a given increase in his stock of a thing, diminishes with every increase in the stock that he already has".
- **In Simple Words:** The more you have of something, the less you want an extra unit of it.

Assumptions (Rules that must be followed for the law to work):

- **Rationality:** The consumer should be a normal person trying to get maximum satisfaction.
- **Cardinal Measurement:** We assume utility can be measured in numbers.
- **Homogeneity:** Every unit consumed must be exactly the same (same size, color, taste).
- **Continuity:** You must consume units one after another without any long breaks.
- **Reasonability:** The units should be of normal size (not too big, not too small).
- **Constancy:** Your income, tastes, and habits must stay the same.
- **Divisibility:** The product can be divided into small parts.
- **Single Want:** The law assumes you are satisfying only one specific want at that time.

Exceptions (Where the law fails):

- **Hobbies:** People who collect stamps or coins get more joy with every new item they add.
- **Misers:** A greedy person gets more satisfaction with every extra rupee they save.
- **Addictions:** For a drunkard, every extra drink might increase their level of intoxication and "perceived" utility.
- **Power:** When people get power, they usually want even more of it.
- **Money:** It is said that the MU of money never becomes zero; we always want more.

7. Criticisms and Significance

Why some people criticize the law:

- It is hard to follow all the strict assumptions (like "homogeneity" or "continuity") in real life.
- Utility is a feeling; you can't really measure it in numbers (Cardinal measurement is unrealistic).

Why the law is important (Significance):

- **For Consumers:** Helps them decide how to spend limited money to get maximum joy.
- **For Government:** Helps in making tax policies (taxing the rich more because the MU of money is lower for them).

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- **Paradox of Values:** Explains why "useless" things like diamonds are expensive (high value-in-exchange) while "essential" things like water are cheap (high value-in-use).
- **Basis of Law of Demand:** This law explains why we buy more only when prices fall.

Short Summary

Utility is the want-satisfying power of a good. It is subjective and relative. We can create utility by changing form, place, or time. The Law of DMU states that as we consume more of a good, the extra satisfaction (MU) we get from each new unit keeps dropping. When MU hits zero, we are at maximum satisfaction (Point of Satiety).

Key Terms

- **Utility:** The capacity of a commodity to satisfy human wants.
- **Marginal Utility (MU):** The addition made to the total utility by the last unit consumed.
- **Total Utility (TU):** The sum total of utilities derived from all units consumed.
- **Point of Satiety:** The point where TU is maximum and MU is zero.
- **Disutility:** Negative marginal utility experienced when consumption continues past the point of satiety.

Exam-Oriented Questions & Answers

- **Q1. Define Utility.**
  - **Ans:** Utility is the capacity of a commodity to satisfy human wants. It is the "want-satisfying power" of a good.
- **Q2. What is the "Point of Satiety"?**
  - **Ans:** It is the stage of consumption where Total Utility (TU) reaches its maximum level and remains constant, while Marginal Utility (MU) becomes zero. This is the point of maximum satisfaction.
- **Q3. State the Law of Diminishing Marginal Utility.**
  - **Ans:** According to Prof. Alfred Marshall, "Other things remaining constant, the additional benefit which a person derives from a given increase in his stock of a thing, diminishes with every increase in the stock that he already has".[1]
- **Q4. Distinguish between Total Utility and Marginal Utility.[2]**
  - **Ans:** Total Utility is the sum of all utilities from all units consumed. Marginal Utility is only the extra utility ga[3]ined from the very last unit consumed.
- **Q5. Mention any two exceptions to the Law of DMU.**

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- **Ans:** 1) Hobbies: Collecting stamps or coins often increases satisfaction with more units. 2) Misers: For a miser, every additional rupee increases his satisfaction rather than decreasing it.

Would you like me to create a table comparing the different types of utility for your quick reference?